

Tab 1

Project Title	Global Seismic Trends: Data-Driven Earthquake Insights.
Skills take away From This Project	• Python for Data Analytics. • Regular Expressions (Regex). • SQL & Relational Database Management. • Streamlit Dashboard Development. • Applied Domain Knowledge – Geoscience & Disaster Management.
Domain	• Disaster Management, • Geoscience • Seismology

Problem Statement:

- Analyze and interpret global earthquake data to identify seismic patterns, trends, and risk zones.
- Build a data-driven system using API-based retrieval, preprocessing, and SQL analytics for meaningful earthquake insights.

Business Use Cases:

- Enable governments, insurers, and researchers to assess earthquake risks, plan disaster management strategies, and support informed decision-making.

- Facilitate data-driven policies for urban safety, infrastructure resilience, and effective emergency response.

Approach:

- Retrieve earthquake data from the USGS API for a defined time range.
- Clean and preprocess the data using Python (Pandas and Regex) for consistency and accuracy.
- Convert timestamps, handle missing values, and derive new analytical columns such as year, month, and depth category.
- Store the cleaned and structured dataset in MySQL for efficient querying and analysis.
- Use SQL to perform detailed analytical exploration, uncovering trends and insights in global seismic activity.

Results:

Expected outcomes by the end of the project:

- A **clean and normalized earthquake dataset** covering the past 5 years.
- A **MySQL database** with all 26 features and derived metrics.
- A series of **analytical SQL queries** providing actionable insights on earthquake trends.
- A **Streamlit dashboard** with interactive visualizations.
- **Documentation** describing key findings, patterns, and recommendations.

Project Evaluation metrics:

- **Data Cleaning Accuracy (35%)** – Quality and correctness of preprocessing, regex extraction, and derived fields.
- **SQL Query Effectiveness (35%)** – Correctness, complexity, and efficiency of analytical queries.
- **Visualization & Dashboard (10%)** – Clarity, interactivity, and analytical depth of Streamlit interface.
- **Documentation & Communication (10%)** – Clarity in explaining insights and methodology.

- **Project Organization (10%)** – Folder structure, naming conventions, and reproducibility.

Dataset retrieval procedure:

- Use the USGS earthquake API:
<https://earthquake.usgs.gov/fdsnws/event/1/query>
- Understand the parameters: starttime, endtime, minmagnitude, and format=geojson.
- Decide the years and months you want to collect data for (e.g., last 5 years).
- Loop through each month and year, create start and end dates for the API request.
- Use requests.get(url, params=parameters) to fetch data for each month.
- Check the response status code and handle any failures.
- Convert the JSON response into a list of dictionaries. Extract data from properties (magnitude, place, status, etc.) and geometry (longitude, latitude, depth).
- Convert timestamp fields from milliseconds to datetime.
- Store all extracted records in a list and then convert it into a Pandas DataFrame.
- Check the DataFrame shape, sample rows, and data types to ensure everything looks correct.

Description of features.

Earthquake Dataset Description (26 Features)

1. id

- **Description:** Unique identifier for each earthquake event.
- **Type:** String
- **Usage:** Serves as the primary key to distinguish events and track them across datasets.

2. time

- **Description:** Exact timestamp when the earthquake occurred.
- **Type:** Datetime
- **Usage:** Useful for time series analysis, identifying trends, or counting events per day, month, or year.

3. updated

- **Description:** Timestamp of the most recent update to the earthquake record.
- **Type:** Datetime
- **Usage:** Helps track revisions and ensures analysis uses the latest data.

4. latitude

- **Description:** Latitude coordinate of the earthquake epicenter.
- **Type:** Float
- **Usage:** Geospatial mapping and plotting earthquake locations on maps.

5. longitude

- **Description:** Longitude coordinate of the earthquake epicenter.
- **Type:** Float
- **Usage:** Geospatial mapping and spatial analysis in combination with latitude.

6. depth_km

- **Description:** Depth of the earthquake in kilometers.
- **Type:** Float

- **Usage:** Analyze shallow vs deep earthquakes; deeper quakes may behave differently than shallow ones.

7. mag

- **Description:** Magnitude of the earthquake.
- **Type:** Float
- **Usage:** Key metric to measure earthquake strength; used for ranking, filtering, and plotting magnitude distributions.

8. magType

- **Description:** Type of magnitude measurement (e.g., mb, ml, mww).
- **Type:** String
- **Usage:** Useful to understand which method was used to compute magnitude and compare consistency.

9. place

- **Description:** Human-readable location description (city, region, country).
- **Type:** String
- **Usage:** Can extract country or region using regex for grouping, filtering, or visualizations.

10. status

- **Description:** Indicates whether the record is reviewed or automatic.
- **Type:** String
- **Usage:** Helps filter high-confidence, reviewed data vs preliminary automated data.

11. tsunami

- **Description:** Indicator (0 or 1) if a tsunami was generated.
- **Type:** Integer
- **Usage:** Useful for analyzing secondary effects of earthquakes or studying tsunami patterns.

12. sig

- **Description:** Significance metric of the earthquake (a measure of impact).
- **Type:** Integer
- **Usage:** Used to prioritize or filter events based on impact/severity.

13. net

- **Description:** Reporting network or agency code (e.g., us, pr).
- **Type:** String
- **Usage:** Helps identify the source of the data; can be used for network-specific analysis.

14. nst

- **Description:** Number of seismic stations that reported the earthquake.
- **Type:** Float
- **Usage:** Measure of data reliability; higher nst usually indicates higher confidence in the event location and magnitude.

15. dmin

- **Description:** Horizontal distance to the nearest station (degrees).
- **Type:** Float

- **Usage:** Used to assess the quality and coverage of the earthquake measurement.

16. rms

- **Description:** Root mean square of residuals.
- **Type:** Float
- **Usage:** Measures how well observed seismic data fits the theoretical model; lower values indicate better fits.

17. gap

- **Description:** Azimuthal gap (in degrees) between stations.
- **Type:** Float
- **Usage:** Quality measure; smaller gaps suggest better station coverage around the epicenter.

18. magError

- **Description:** Uncertainty in the magnitude measurement.
- **Type:** Float
- **Usage:** Indicates confidence in the reported magnitude; used for filtering high-confidence events.

19. depthError

- **Description:** Uncertainty in depth measurement.
- **Type:** Float
- **Usage:** Helps assess reliability of the depth value; lower error = more reliable.

20. magNst

- **Description:** Number of stations contributing specifically to magnitude calculation.
- **Type:** Float
- **Usage:** Another reliability metric; can be used to filter events with low data support.

21. locationSource

- **Description:** Network or agency reporting the location.
- **Type:** String
- **Usage:** Can track which agency's data was used for epicenter calculations.

22. magSource

- **Description:** Network or agency reporting magnitude.
- **Type:** String
- **Usage:** Useful for verifying consistency between agencies' magnitude measurements.

23. types

- **Description:** Types of associated data (e.g., origin, phase-data, dyfi, shakemap).
- **Type:** String
- **Usage:** Helps filter for certain types of earthquake data; useful for research purposes.

24. ids

- **Description:** Comma-separated list of all IDs associated with the event.

- **Type:** String
- **Usage:** References multiple identifiers for the same earthquake; useful for cross-referencing.

25. sources

- **Description:** Comma-separated list of reporting sources for this event.
- **Type:** String
- **Usage:** Identifies agencies that reported this event; can be used for reliability assessment.



- **Description:** Type of seismic event (usually “earthquake”).
- **Type:** String
- **Usage:** Can filter different event types if dataset includes other phenomena like explosions or quarry blasts.

Data Preparation

Objective: Clean, transform, and store your API data for analysis.

1. Load Data

- Load the dataset directly from API or CSV into Pandas.
- Convert date/time fields (**time**, **updated**) to **datetime** objects.

2. Clean Text Fields

- Use Regex to extract **country** from **place**.
- Normalize **alert** field (if exists) to lowercase.
- Ensure all string fields (**magType**, **status**, **type**, **net**, **sources**, **types**) are clean.

3. Clean Numeric Fields

- Convert **mag**, **depth_km**, **nst**, **dmin**, **rms**, **gap**, **magError**, **depthError**, **magNst**, **sig** to numeric.
- Fill missing numeric values with 0 or median if needed.

4. Add Derived Columns

- Year, month, day, day_of_week from **time**.
- Shallow/deep earthquake flag based on **depth_km**.
- Strong/destructive flag based on **mag** thresholds.

5. Store Data in MySQL

- Create table **earthquakes** with all 26 features and derived columns.
- Insert cleaned DataFrame using **SQLAlchemy**.

Analyst Tasks

Objective: Leverage SQL to generate insights using **all 26 features**.

Magnitude & Depth

1. Top 10 strongest earthquakes (**mag**).
2. Top 10 deepest earthquakes (**depth_km**).
3. Shallow earthquakes < 50 km and mag > 7.5.
4. Average depth per continent.
5. Average magnitude per magnitude type (**magType**).

Time Analysis

6. Year with most earthquakes.
7. Month with highest number of earthquakes.
8. Day of week with most earthquakes.
9. Count of earthquakes per hour of day.
10. Most active reporting network (**net**).

Casualties & Economic Loss

11. Top 5 places with highest casualties.
12. Total estimated economic loss per continent.
13. Average economic loss by alert level.

Event Type & Quality Metrics

14. Count of reviewed vs automatic earthquakes (**status**).
15. Count by earthquake type (**type**).
16. Number of earthquakes by data type (**types**).
17. Average RMS and gap per continent.
18. Events with high station coverage (**nst > threshold**).

Tsunamis & Alerts

19. Number of tsunamis triggered per year.
20. Count earthquakes by alert levels (**red**, **orange**, etc.).

Seismic Pattern & Trends Analysis.

21. Find the top 5 countries with the highest average magnitude of earthquakes in the past 10 years

22. Find countries that have experienced both shallow and deep earthquakes within the same month.

23. Compute the **year-over-year growth rate** in the total number of earthquakes globally.

24. List the **3 most seismically active regions** by combining both frequency and average magnitude.

Depth, Location & Distance-Based Analysis.

25. For each country, calculate the average depth of earthquakes within $\pm 5^\circ$ latitude range of the equator.

26. Identify countries having the highest ratio of shallow to deep earthquakes.

27. Find the average magnitude difference between earthquakes with tsunami alerts and those without.

28. Using the gap and rms columns, identify events with the lowest data reliability (highest average error margins).

29. Find pairs of consecutive earthquakes (by time) that occurred within 50 km of each other and within 1 hour.

30. Determine the regions with the highest frequency of deep-focus earthquakes (depth > 300 km).

Deliverables

1. **MySQL Database** – fully cleaned, stored with all features.
2. **SQL Queries** – covering magnitude, depth, casualties, time trends, and quality metrics.
3. **Python Analysis** – full exploratory analysis using Pandas, Regex,

4. **Streamlit Dashboard** – interactive, multi-filtered visualization.
5. **Documentation** – explaining each feature, analysis, and insights.

Evaluation Metrics:

Accuracy and effectiveness of data cleaning and preprocessing using Python and Regex. Correctness of SQL queries, quality of insights, and clarity of visualizations in Streamlit dashboards. Project organization, interactivity, and the ability to communicate actionable findings clearly will also be assessed.

Streamlit recording (Tamil)	 Streamlit - Tamil
Streamlit recording (English)	 Special session for STREAMLIT
Streamlit Reference doc	Streamlit API reference

Project Excellence Series for Python & SQL Project (Eng)	Project Excellence Series: Guided Learn...
GitHub Reference	How to Use GitHub.pptx
Capstone Explanation Guideline	Capstone Explanation Guideline
Project Live Evaluation Metrics	Project Live Evaluation
Project Orientation session link - Tamil recording	File Project Orientation Session : Global Seis...

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.

Timing: Monday-Saturday (3:30PM to 4:30PM)

Booking link :  **Project Doubt Session [DS/AIML]**

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.

Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timing:

For DS and AIML

Monday-Saturday (05:30PM to 07:00PM)

- **Booking link :** <https://forms.gle/1m2Gsro41fLtZurRA>

Evaluation Metrics : [Project Live Evaluation Metrics](#)

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